

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 23
Design of FIR Filters Using Frequency Sampling

III B.Tech EEE — Core Exam Preparation Guide

1 Core Concept of Frequency Sampling

1.1 Basic Principle

Instead of truncating an infinite impulse response in the time domain (windowing), the **Frequency Sampling Method** directly discretizes the desired continuous frequency response $H_d(e^{j\omega})$ at N uniformly spaced points on the unit circle:

$$\omega_k = \frac{2\pi k}{N}, \quad k = 0, 1, \dots, N-1 \quad (1)$$

The discrete frequency samples are:

$$H[k] = H_d(e^{j\omega})|_{\omega=\frac{2\pi k}{N}} = |H[k]|e^{j\theta_k} \quad (2)$$

The causal FIR impulse response $h[n]$ ($0 \leq n \leq N-1$) is computed via the N -point **Inverse Discrete Fourier Transform (IDFT)**:

$$h[n] = \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{j\frac{2\pi kn}{N}}, \quad 0 \leq n \leq N-1 \quad (3)$$

By definition, the reconstructed filter response $H(e^{j\omega})$ passes **exactly** through $H[k]$ at all N grid frequencies:

$$H\left(e^{j\frac{2\pi k}{N}}\right) \equiv H[k], \quad \forall k \in [0, N-1] \quad (4)$$

2 Linear-Phase Constraints on Samples $H[k]$

For the synthesized impulse response $h[n]$ to be real-valued and satisfy linear phase ($h[n] = h[N-1-n]$), the frequency samples must satisfy:

1. Magnitude Symmetry (Even Symmetry):

$$|H[N-k]| = |H[k]|, \quad k = 1, 2, \dots, N-1 \quad (5)$$

2. Phase Angle Assignment (Strict Linear Phase):

$$\theta_k = \angle H[k] = -\tau\omega_k = -\left(\frac{N-1}{2}\right) \frac{2\pi k}{N} \quad (6)$$

3. DC Bin Property:

$$\theta_0 = 0 \implies H[0] \in \mathbb{R}^+ \quad (\text{Strictly real and non-negative}) \quad (7)$$

4. **Nyquist Bin Property (for N even):** At $k = N/2$ ($\omega = \pi$), Type 2 filters possess an obligatory zero:

$$\mathbf{H}[\mathbf{N}/2] = \mathbf{0} \quad (8)$$

3 Real-Valued IDFT Synthesis Formulas (Master Formulas)

By combining complex conjugate pairs $H[k]$ and $H[N - k]$, all complex exponentials collapse into real trigonometric cosines:

Table 1: Operational Closed-Form Frequency Sampling Formulas

Filter Type	Formula for Impulse Response $h[n]$ ($0 \leq n \leq N - 1$)	Center Delay τ
Type 1 (N Odd)	$h[n] = \frac{1}{N} \left\{ H[0] + 2 \sum_{k=1}^{(N-1)/2} H[k] \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right] \right\}$	$\tau = \frac{N-1}{2}$ (Integer)
Type 2 (N Even)	$h[n] = \frac{1}{N} \left\{ H[0] + 2 \sum_{k=1}^{N/2-1} H[k] \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right] \right\}$	$\tau = \frac{N-1}{2}$ (Half-Integer, $H[N/2] = 0$)

4 Transition Sample Optimization (Sidelobe Control)

- **The Ringing Problem:** Naive brick-wall frequency sampling (setting passband samples to 1 and stopband to 0) causes severe inter-sample ringing between bins, resulting in poor peak stopband attenuation of only ****−16 dB****!
- **The Solution (Rabiner–Gold Technique):** Make the samples in the transition band unconstrained design variables ($T_1, T_2 \in (0, 1)$).
 - With **1 transition sample** ($T_1 \approx 0.38$): Peak stopband attenuation jumps to ****−45 dB**** (+29 dB improvement!).
 - With **2 transition samples** ($T_1 \approx 0.59, T_2 \approx 0.11$): Stopband attenuation reaches ****−75 dB****.

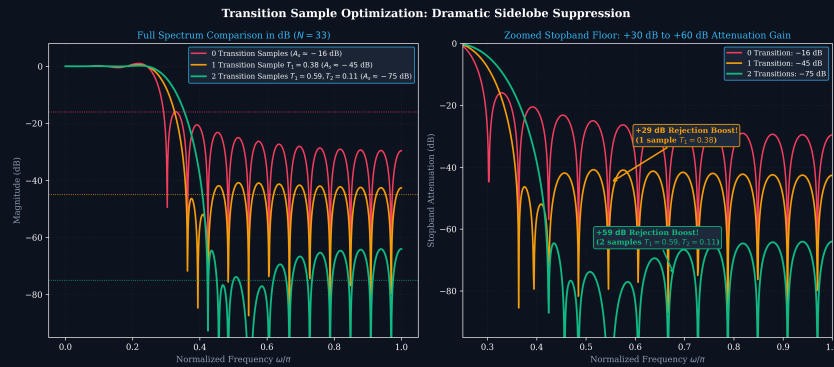


Figure 1: Dramatic sidelobe suppression achieved by introducing transition sample $T_1 = 0.38$ between passband and stopband.

5 Standard Solved Exam Example

Problem: Design a 7-point ($N = 7$) linear-phase FIR lowpass filter having the following desired frequency response:

$$H_d(e^{j\omega}) = \begin{cases} 1, & 0 \leq |\omega| \leq \frac{2\pi}{7} \\ 0, & \frac{4\pi}{7} \leq |\omega| \leq \pi \end{cases} \quad (9)$$

using the frequency sampling method. Determine the impulse response $h[n]$ for $n = 0, 1, 2, 3, 4, 5, 6$.

Solution:

Step 1: Determine Sampling Frequencies and Samples For $N = 7$, grid frequencies are $\omega_k = \frac{2\pi k}{7}$:

- $k = 0$: $\omega_0 = 0 \implies |H[0]| = 1$
- $k = 1$: $\omega_1 = \frac{2\pi}{7} \leq \frac{2\pi}{7} \implies |H[1]| = 1$
- $k = 2$: $\omega_2 = \frac{4\pi}{7} \geq \frac{4\pi}{7} \implies |H[2]| = 0$
- $k = 3$: $\omega_3 = \frac{6\pi}{7} \geq \frac{4\pi}{7} \implies |H[3]| = 0$

By Hermitian symmetry: $|H[6]| = |H[1]| = 1$, $|H[5]| = |H[2]| = 0$, $|H[4]| = |H[3]| = 0$.

Step 2: Apply the Type 1 Real IDFT Formula Since $N = 7$ is odd, $(N - 1)/2 = 3$, and the center delay is $\tau = 3$:

$$h[n] = \frac{1}{7} \left\{ H[0] + 2 \sum_{k=1}^3 |H[k]| \cos \left[\frac{2\pi k}{7} (n - 3) \right] \right\} \quad (10)$$

Since $|H[2]| = |H[3]| = 0$, only the $k = 1$ term survives!

$$h[n] = \frac{1}{7} \left\{ 1 + 2 \cos \left[\frac{2\pi}{7} (n - 3) \right] \right\}, \quad 0 \leq n \leq 6 \quad (11)$$

Step 3: Pointwise Tap Evaluation

- **Center Tap ($n = 3$):**

$$h[3] = \frac{1 + 2 \cos(0)}{7} = \frac{1 + 2(1)}{7} = \frac{3}{7} \approx \mathbf{0.4286} \quad (12)$$

- **Adjacent Taps ($n = 2$ and $n = 4$):** For $n = 2$, $n - 3 = -1$:

$$h[2] = h[4] = \frac{1 + 2 \cos(2\pi/7)}{7} = \frac{1 + 2 \cos(51.43^\circ)}{7} \approx \frac{1 + 2(0.6235)}{7} = \frac{2.2470}{7} \approx \mathbf{0.3210} \quad (13)$$

- **Next Taps ($n = 1$ and $n = 5$):** For $n = 1$, $n - 3 = -2$:

$$h[1] = h[5] = \frac{1 + 2 \cos(4\pi/7)}{7} = \frac{1 + 2 \cos(102.86^\circ)}{7} \approx \frac{1 + 2(-0.2225)}{7} = \frac{0.5549}{7} \approx \mathbf{0.0793} \quad (14)$$

- **Outer Taps ($n = 0$ and $n = 6$):** For $n = 0$, $n - 3 = -3$:

$$h[0] = h[6] = \frac{1 + 2 \cos(6\pi/7)}{7} = \frac{1 + 2 \cos(154.29^\circ)}{7} \approx \frac{1 + 2(-0.9010)}{7} = \frac{-0.8019}{7} \approx \mathbf{-0.1146} \quad (15)$$

Step 4: Final Sequence and Symmetry Verification

$$h[n] = \{-0.1146, 0.0793, 0.3210, \underset{\uparrow}{0.4286}, 0.3210, 0.0793, -0.1146\} \quad (16)$$

Symmetry check: $h[n] = h[6 - n]$ holds exactly. Constant group delay $\tau = 3$ samples.

6 Essential Exam Checklist & Traps

- ☐ **The Missing Factor of 2:** Always write $2 \sum |H[k]| \cos(\dots)$. Forgetting the factor of 2 is the most common exam error!
- ☐ **Argument of Cosine:** The argument is $\frac{2\pi k}{N}(n - \tau)$, where $\tau = (N - 1)/2$.
- ☐ **Even Length N Boundary Condition:** When N is even, you **must set** $H[N/2] = 0$ because Type 2 filters have an obligatory zero at $\omega = \pi$.
- ☐ **Exact Values at Grid Points:** At $\omega = \omega_k = \frac{2\pi k}{N}$, the filter response is strictly equal to $H[k]$ with zero approximation error.